

Bash Scripting Challenge: Log Analyzer and Report Generator

❖ Task 1: Input and Validation

- Accept the path to a log file as a command-line argument
- Exit with a clear error message if no argument is provided
- Exit with a clear message if the file doesn't exist
 - `#!/bin/bash`
 - `# Check if argument is provided`
 - `if [ $# -eq 0 ]; then`
 - `echo "Error: No log file path provided. Usage: $0`
`<log_file_path>"`
 - `exit 1`
 - `fi`
 - `# Check if file exists`
 - `if [ ! -f "$1" ]; then`
 - `echo "Error: Log file '$1' does not exist."`
 - `exit 1`
 - `fi`
 - `echo "Log file '$1' validated successfully."`

❖ Task 2: Error Count

- Count the total number of lines containing the keyword ERROR or Failed
- Print the total error count to the console
 - `ERROR_COUNT=$(grep -Ei "ERROR|Failed" "$LOG_FILE" | wc -l)`
 - `echo "Total ERROR/Failed count: $ERROR_COUNT"`

➤ Task 3: Critical Events

- Search for lines containing the keyword Critical
- Print that line along with their line number
 - `echo ""`
 - `echo "--- Critical Events ---"`
 - `CRITICAL_LINES=$(grep -n "CRITICAL" "$LOG_FILE")`
 -
 - `if [ -z "$CRITICAL_LINES" ]; then`
 - `echo "No critical events found."`

- else
- echo "\$CRITICAL_LINES" | while read line; do
- LINE_NUM=\$(echo "\$line" | cut -d: -f1)
- MESSAGE=\$(echo "\$line" | cut -d: -f2-)
- echo "Line \$LINE_NUM: \$MESSAGE"
- done
- Fi

❖ Task 4: Top Error Messages

- Extract all lines containing ERROR
- Identify them with their occurrence count, sorted in descending order
 - echo ""
 - echo "--- Top 5 Error Messages ---"
 -
 - TOP_ERRORS=\$(grep "ERROR" "\$LOG_FILE" \
 - | awk '{ \$1=\$2=\$3=""; print }' \
 - | sort \
 - | uniq -c \
 - | sort -rn \
 - | head -5)
 -
 - if [-z "\$TOP_ERRORS"]; then
 - echo "No ERROR messages found."
 - else
 - echo "\$TOP_ERRORS"
 - Fi

❖ Task 5: Summary Report

- Generate a summary report to a text file named log_report_<date>.text. the should include:
 - Date of analysis
 - Log file name
 - Total lines Processed
 - Total error count
 - Top 5 error message with their occurrence count
 - List of critical events with line numbers

```
ubuntu@ip-172-31-47-149:~$ cat not_exist_file.txt 2>> sample.log
ubuntu@ip-172-31-47-149:~$ touch demofile
ubuntu@ip-172-31-47-149:~$ chmod 000 demofile
ubuntu@ip-172-31-47-149:~$ cat demofile 2>> sample.log
ubuntu@ip-172-31-47-149:~$ su
Password:
ubuntu@ip-172-31-47-149:~$ cd /
ubuntu@ip-172-31-47-149:/$ cd
ubuntu@ip-172-31-47-149:~$ sudo systemctl start fake_service 2>> sample.log
ubuntu@ip-172-31-47-149:~$ ./log_analyzer.sh sample.log
🔍 Analyzing log file: sample.log
-----
Total ERROR/Failed count: 1

--- Critical Events ---
No critical events found.

--- Top 5 Error Messages ---
No ERROR messages found.

Report generated: log_report_2026-02-24.txt
ubuntu@ip-172-31-47-149:~$
```

❖ What I Learned

- Analyzing log files with commands like grep, awk
- Writing reports and archiving them.